Sequencing-Depth Robustness of Cell-Free DNA Fragmentomic Features for Breast Cancer Classification

Nelly Abualkheir

Brandeis University

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Abstract

Cell-free DNA (cfDNA) fragmentomics provides information beyond sequence variants by examining characteristics such as fragment length and genome-wide fragmentation patterns. These patterns are influenced by nucleosome organization, chromatin accessibility, and the tissues contributing DNA to plasma, making fragmentomics a promising approach for non-invasive cancer detection. However, genome-wide fragmentomic analysis can depend strongly on sequencing depth, which may affect both analytical stability and classification performance. This project evaluated how simulated reductions in sequencing depth influence cfDNA fragmentomic features and breast cancer classification.

Publicly available cfDNA whole-genome fragmentation data were analyzed from 92 plasma samples, including 46 breast cancer samples and 46 healthy controls. Global fragment-size features and genome-wide regional short-to-long fragment ratios were calculated and evaluated separately and in combination. Logistic regression models were assessed using repeated stratified cross-validation. Sequencing depth was then computationally reduced to 20, 10, 5, 2, and 1 million fragments per sample to evaluate the stability of regional profiles and the resulting change in cancer classification performance.

Global fragment-size features alone showed limited discrimination between breast cancer and healthy samples, with a ROC-AUC of approximately 0.65. In contrast, genome-wide regional fragmentation patterns achieved a ROC-AUC of approximately 0.92 at full depth. Regional profiles remained highly correlated with full-depth profiles at 20 million fragments but progressively lost stability as sequencing depth decreased. Classification followed the same trend, declining from a full-depth ROC-AUC of approximately 0.92 to 0.87 at 20 million, 0.82 at 10 million, 0.80 at 5 million, 0.66 at 2 million, and 0.62 at 1 million fragments. These results suggest that regional cfDNA fragmentation contains diagnostically informative structure that is not captured by global fragment-size summaries, but sufficient sequencing coverage is required to preserve this information.

Part 1. Proposal & Scientific Question

Scientific Question

How does simulated reduction in sequencing depth affect the stability of global and genome-wide cfDNA fragmentomic features, and which features retain sufficient diagnostic performance for cancer-versus-healthy classification at low fragment counts?

Cell-free DNA has become an important source of molecular information for liquid biopsy because DNA released into the circulation can be analyzed using a blood sample. Traditional circulating tumor DNA approaches have focused heavily on tumor-associated mutations and copy-number changes. However, tumor-derived DNA represents only a fraction of the total cfDNA present in plasma, particularly when tumor burden is low. This creates an important limitation for early cancer detection, where the number of molecules carrying detectable tumor-specific alterations may be very small.

Fragmentomics approaches use a different layer of information. Rather than asking only which variants are present, fragmentomics examines how DNA is fragmented in plasma. cfDNA fragmentation is not random. Fragment length, genomic position, fragment ends, and regional fragmentation patterns are influenced by chromatin structure and nucleosome organization in the cells from which the DNA originated. Cancer can alter these patterns through changes in chromatin organization, cell turnover, and the mixture of tissues contributing DNA to plasma. Previous work has demonstrated that genome-wide fragmentation patterns can distinguish individuals with cancer from healthy individuals, supporting the potential of fragmentomics as a complementary liquid-biopsy signal (Cristiano et al., 2019).

Q1. Why is this problem important?

One practical challenge is that fragmentomic measurements depend on sequencing coverage. A model may perform well using high-depth whole-genome sequencing, but that performance does not automatically mean the same biological signal will remain measurable when fewer fragments are available. This distinction is important for clinical translation. Sequencing depth affects cost, computational requirements, turnaround time, and the amount of usable information available from each sample. A diagnostic approach that requires extremely high sequencing depth may be difficult to scale, while reducing depth too far may remove the regional signal that makes fragmentomics useful.

The important question is therefore not simply whether fragmentomics can classify cancer. It is whether the fragmentomic signal remains stable enough to be informative as sequencing depth decreases.

Q2. How could the findings translate to biology or patient care?

Understanding depth robustness can help identify which types of fragmentomic information are most useful for assay development. If simple global features remain stable but provide little diagnostic information, increasing their technical precision would not necessarily improve cancer classification. In contrast, if regional fragmentation patterns provide strong discrimination but become unstable below a certain depth, sequencing depth becomes an important analytical design parameter.

This project therefore connects a computational question with a practical clinical problem: how much fragment-level information is needed before a genome-wide fragmentation profile becomes sufficiently stable to support cancer classification?

The original hypothesis was that regional fragmentation profiles would provide stronger cancer-versus-healthy discrimination than global fragment-size features because they preserve genomic context, but that their performance would progressively decrease as sequencing depth was reduced. I also expected moderate reductions in depth to retain useful information before a more substantial loss of performance occurred at very low fragment counts.

Part 2. Dataset Selection

Q3. Data Source and Dataset Size

The analysis used publicly available cfDNA fragmentation data obtained through FinaleDB, the FragmentatIoN AnaLysis of cEll-free DNA DataBase. FinaleDB was developed to provide uniformly processed cfDNA whole-genome sequencing datasets while removing sequence information that could reveal patient genotypes. The database contains paired-end cfDNA datasets representing multiple diseases and physiological conditions and provides fragment-level information suitable for fragmentomic analyses (Zheng et al., 2021).

For this study, I selected a balanced cohort of 92 plasma cfDNA samples, consisting of 46 breast cancer samples and 46 healthy controls.

Study group	Number of samples
Breast cancer	46
Healthy	46
Total	92

Each sample contained genome-aligned fragment information from which fragment size and genomic location could be analyzed. Observed sequencing depth varied substantially among samples, ranging from approximately 22.9 million to 148.5 million fragments.

Q4. Why This Dataset Was Appropriate

This dataset was well suited to the scientific question for several reasons. First, it contained fragment-level information rather than only previously calculated summary features. This allowed global fragment-size distributions and regional genomic features to be calculated using the same underlying samples.

Second, the cohort was balanced between breast cancer and healthy controls. The equal group size reduced concerns that classification performance would simply reflect a dominant class and made accuracy, F1 score, balanced accuracy, and ROC-AUC easier to interpret together.

Third, the available sequencing depth was substantially higher than the lowest depths evaluated in this project. This made it possible to treat the observed data as a full-depth reference and computationally simulate lower fragment counts. Each sample could therefore be compared against its own original fragmentation profile, which reduced biological variability when evaluating sequencing-depth robustness.

Finally, FinaleDB applies standardized processing to public cfDNA WGS datasets. This is useful because fragmentomic measurements can be affected by read length, alignment strategy, sequencing technology, and other technical factors. The dataset was therefore appropriate for testing whether regional and global cfDNA fragmentation signals behave differently as the number of available fragments decreases (Zheng et al., 2021).

Part 3. Methods Overview

Q5. Bioinformatics Workflow

The analysis was organized as a reproducible workflow beginning with public fragment-level cfDNA data and ending with depth-dependent profile stability and classification performance. The workflow below separates data QC and feature generation from the two main analytical branches: full-depth cancer classification and sequencing-depth robustness testing.

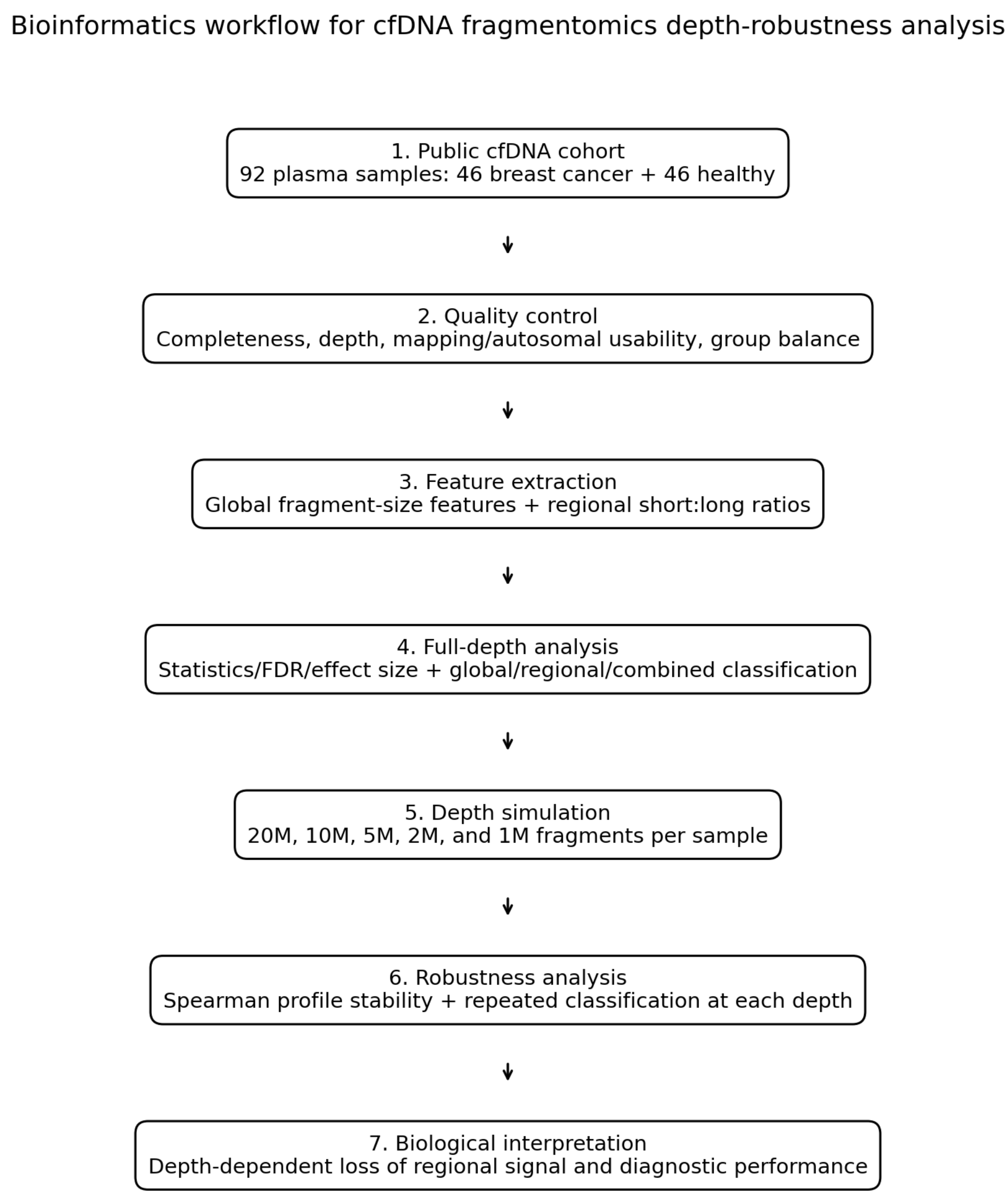

Figure 1. Bioinformatics workflow for the cfDNA fragmentomics depth-robustness analysis.

Q6. Computational Methodology

The workflow was implemented in Python using Google Colab. Data manipulation and numerical operations were performed primarily with pandas and NumPy, statistical analyses with SciPy and statsmodels, machine-learning analyses with scikit-learn, and visualization with Matplotlib.

Initial QC included sample counts, group balance, missing values, sequencing depth, mapping-related metrics, and usable autosomal fragment fractions. The final cohort contained 46 breast cancer and 46 healthy samples with no missing sample-level QC values.

Global fragmentation features were derived from fragment-length distributions. These included mean fragment length, median length, standard deviation of fragment length, proportions of fragments below 150 bp, from 100–150 bp, and from 151–220 bp, a short-to-long fragment ratio, and fragment-length entropy.

Genome-wide regional fragmentation was represented using short-to-long fragment ratios calculated across genomic bins. Unlike a global ratio, these features preserve information about where in the genome fragmentation differences occur.

Group differences in global features were evaluated using the Mann–Whitney U test. Because multiple global features were tested, p-values were corrected using the Benjamini–Hochberg false discovery rate procedure. Rank-biserial correlation was calculated as an effect-size measure.

For classification, a logistic-regression model with L2 regularization was implemented within a StandardScaler pipeline. Performance was evaluated using 5-fold repeated stratified cross-validation with 10 repeats, giving 50 train/test evaluations while maintaining the breast cancer/healthy balance within each fold. ROC-AUC was the primary classification metric, with accuracy, F1 score, and balanced accuracy reported as supporting measures.

Sequencing-depth robustness was evaluated by computational downsampling. Fragment counts were simulated at 20 million, 10 million, 5 million, 2 million, and 1 million fragments per sample. The sampling process preserved the expected relative representation of short and long fragments. In a validation check, the expected retained fraction was 0.2562, while observed retained fractions were 0.2562 for short fragments and 0.2561 for long fragments.

Regional profiles generated after downsampling were compared with their corresponding full-depth profiles using Spearman correlation. This quantified how closely the genome-wide pattern was retained as fragments were removed. For classification at reduced depth, 20 independent downsampling replicates were generated at each target depth. Each replicate was evaluated using 5-fold stratified cross-validation with the same standardized L2-regularized logistic-regression pipeline. Sequencing depth was therefore evaluated at two levels: preservation of the underlying fragmentation profile and preservation of diagnostic classification performance.

Part 4. Results & Visualization

Q7. Quality Control

All 92 samples were successfully incorporated into the final cohort, with 46 breast cancer and 46 healthy controls. No missing sample-level QC values were observed.

Sequencing depth showed substantial variability across the cohort. Breast cancer samples had a median depth of approximately 42.8 million fragments, while healthy samples had a median of approximately 42.9 million fragments. Although several high-depth outliers were present, particularly among healthy samples, the distributions did not differ significantly by group (Mann–Whitney $U = 986$, $p = 0.577$). This indicated that cancer classification was unlikely to be explained simply by systematic differences in sequencing depth.

During regional downsampling, one genomic feature, chr9_45000000, produced missing values and was removed from the stability analysis. This reduced the regional feature set from 570 to 569 stable regions, after which no missing regional values remained.

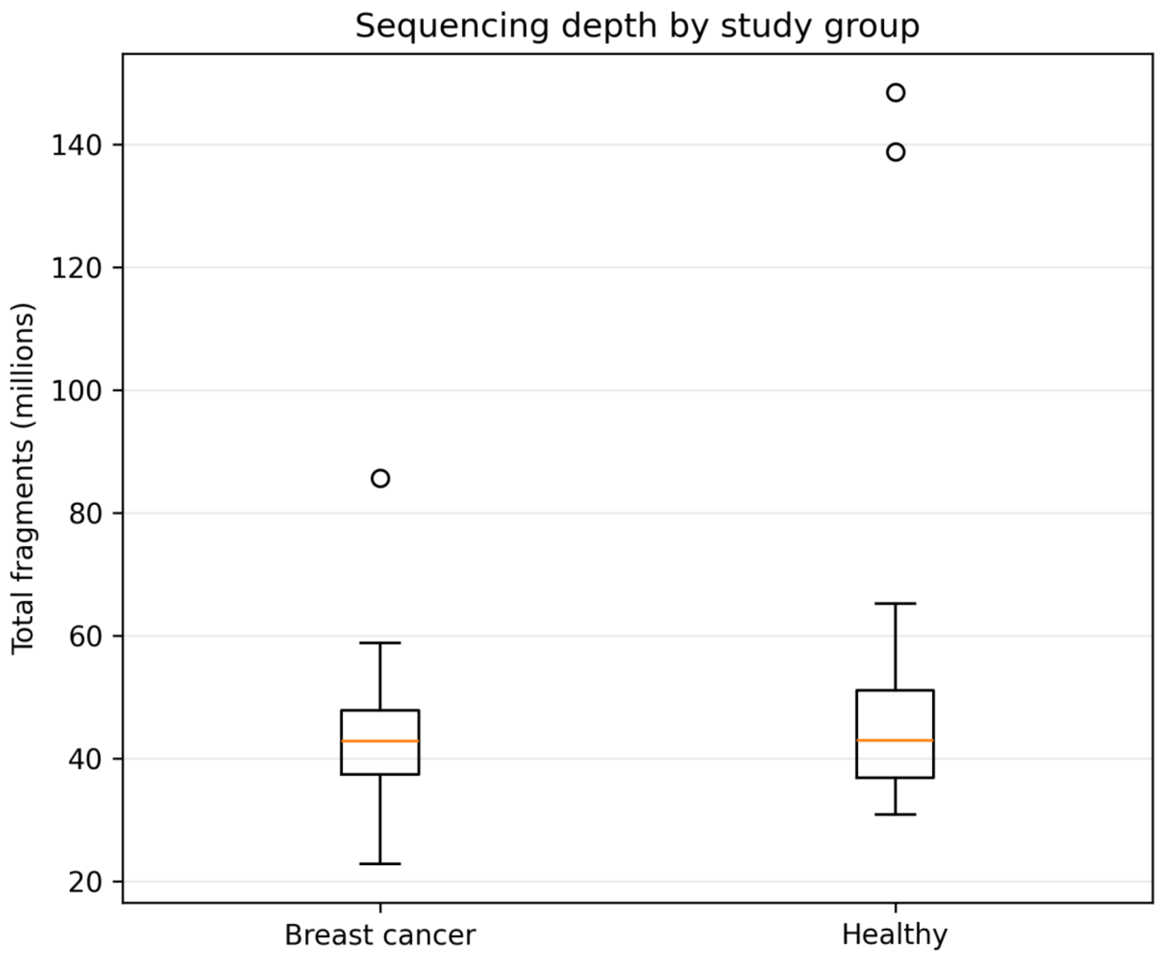

Figure 2. Sequencing depth by study group. Breast cancer and healthy controls had similar median fragment counts; the between-group difference was not significant (Mann–Whitney U = 986, p = 0.577).

Q8–Q9. Primary Results and Statistical Evaluation

Global fragment-size features showed relatively weak group separation. Fragment entropy and fragment-length standard deviation produced the smallest unadjusted p-values, at $p = 0.0082$ and $p = 0.0126$, respectively. However, after Benjamini–Hochberg correction, both had an FDR-adjusted p-value of approximately 0.0504 and therefore did not meet the predefined FDR < 0.05 threshold. Their rank-biserial correlations were approximately −0.32 and −0.30, suggesting modest effects rather than strong separation.

This pattern was consistent with the classification results. Global features alone achieved a ROC-AUC of approximately 0.646 ± 0.114. In contrast, the genome-wide regional fragmentation model achieved approximately 0.917 ± 0.056, and combining regional and global features produced approximately 0.921 ± 0.057. The small improvement from 0.917 to 0.921 suggests that most of the discriminative information was contained in the regional profiles rather than the global summary statistics.

At 20 million fragments, the median Spearman correlation with the full-depth profile was approximately 0.948. This decreased to 0.874 at 10M, 0.774 at 5M, 0.616 at 2M, and 0.493 at 1M. Therefore, the regional profile remained highly similar to the original profile at 20M but progressively deteriorated as sequencing depth decreased.

Simulated depth	Mean ROC-AUC
Full depth	~0.917
20M	0.871
10M	0.822
5M	0.803
2M	0.655
1M	0.619

Classification followed the same depth-dependent trend. The model retained relatively strong discrimination at 20M fragments and remained informative at 10M and 5M. Performance dropped more sharply below 5M, reaching approximately 0.66 ROC-AUC at 2M and 0.62 at 1M. Together, the profile-stability and classification analyses show that loss of predictive performance was accompanied by loss of the underlying regional fragmentation pattern.

The primary quantitative results are summarized below using the final model-comparison table, regional-profile stability summary, depth-classification table, and corresponding diagnostic figures.

Table 2. Full-depth breast cancer classification performance by fragmentomic feature set.

Feature set	ROC-AUC (mean ± SD)
Global	0.646 ± 0.114
Regional	0.917 ± 0.056
Combined	0.921 ± 0.057

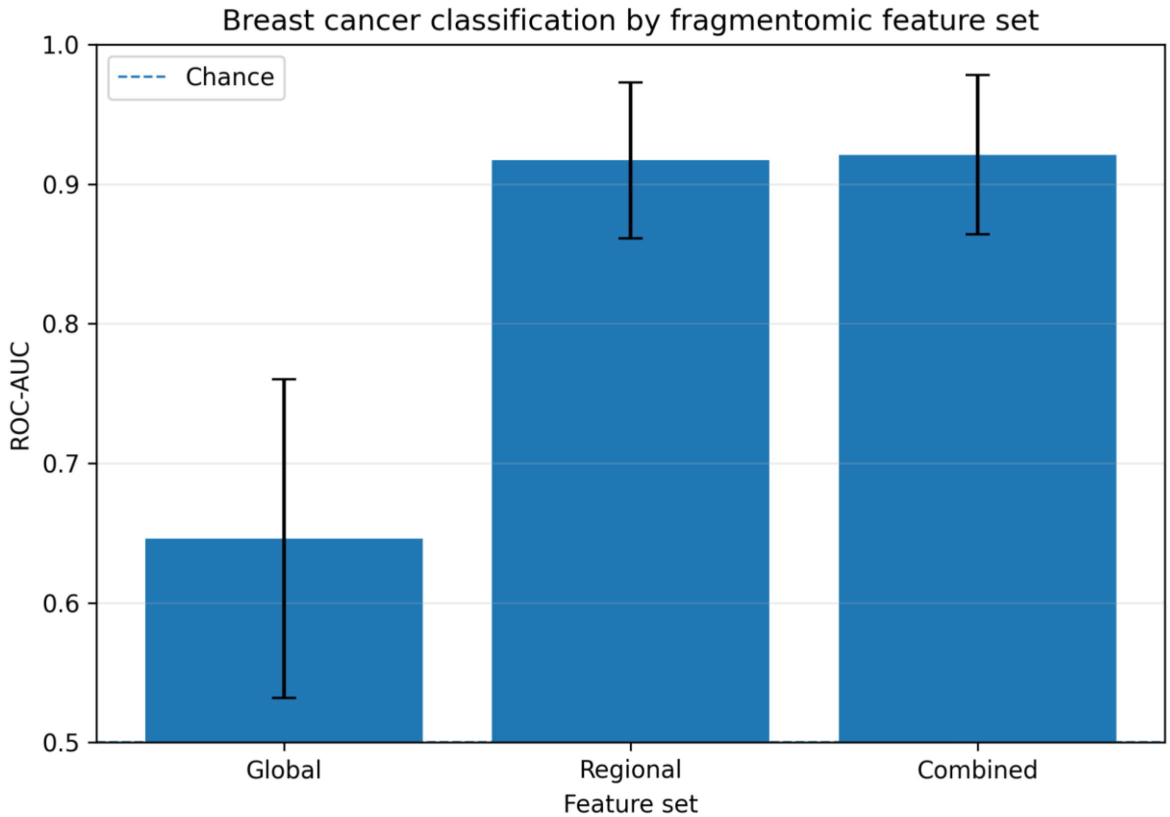

Figure 3. Breast cancer classification by fragmentomic feature set. Regional and combined models substantially outperformed global fragment-size features; error bars show cross-validation variability.

Table 3. Regional-profile stability relative to the full-depth reference across simulated sequencing depths.

Target depth	Spearman ρ (mean ± SD)	Median ρ
20M	0.947 ± 0.018	0.948
10M	0.873 ± 0.034	0.874
5M	0.776 ± 0.052	0.774
2M	0.621 ± 0.070	0.616
1M	0.500 ± 0.076	0.493

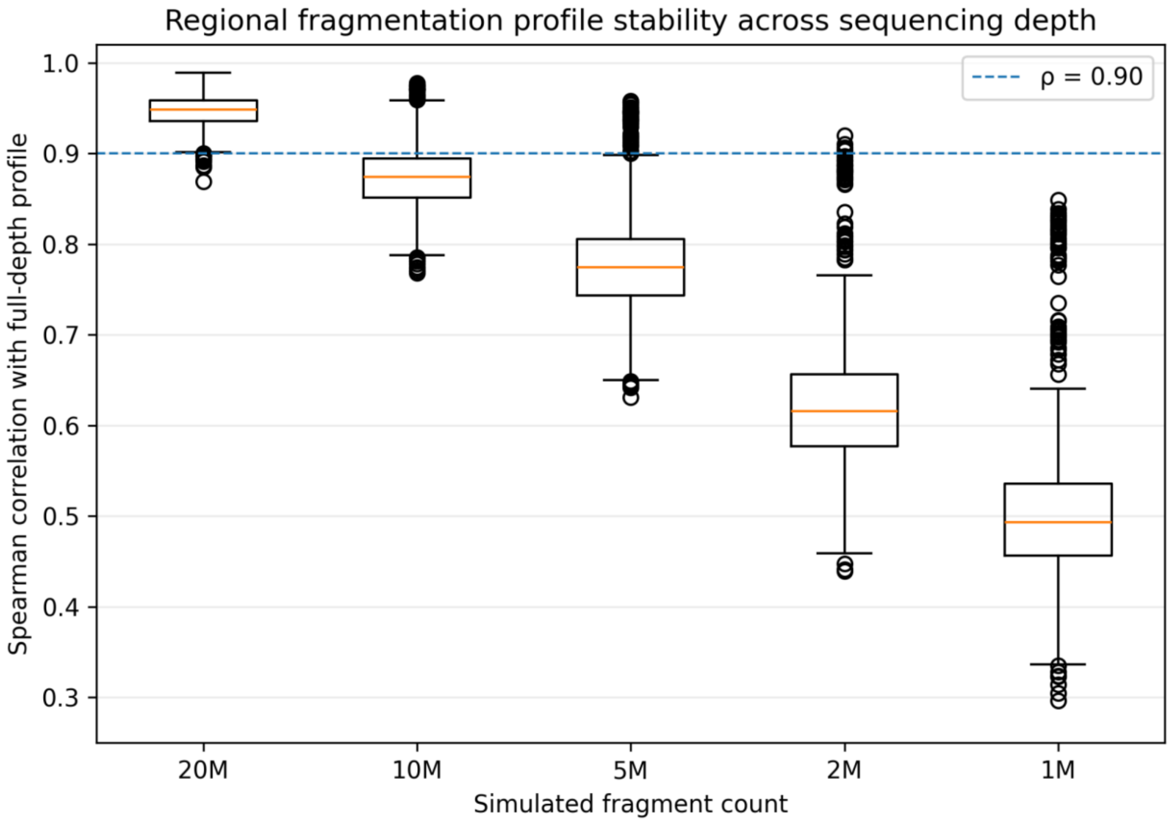

Figure 4. Regional fragmentation profile stability across sequencing depth. The dashed line marks ρ = 0.90; profile concordance progressively declined as fragment counts were reduced.

Table 4. Breast cancer classification performance across simulated sequencing depths.

Depth	ROC-AUC (mean ± SD)	Accuracy	F1	Balanced accuracy
20M	0.871 ± 0.030	0.799	0.788	0.801
10M	0.822 ± 0.037	0.740	0.726	0.742
5M	0.803 ± 0.054	0.722	0.710	0.725
2M	0.655 ± 0.079	0.608	0.597	0.609
1M	0.619 ± 0.072	0.588	0.573	0.588

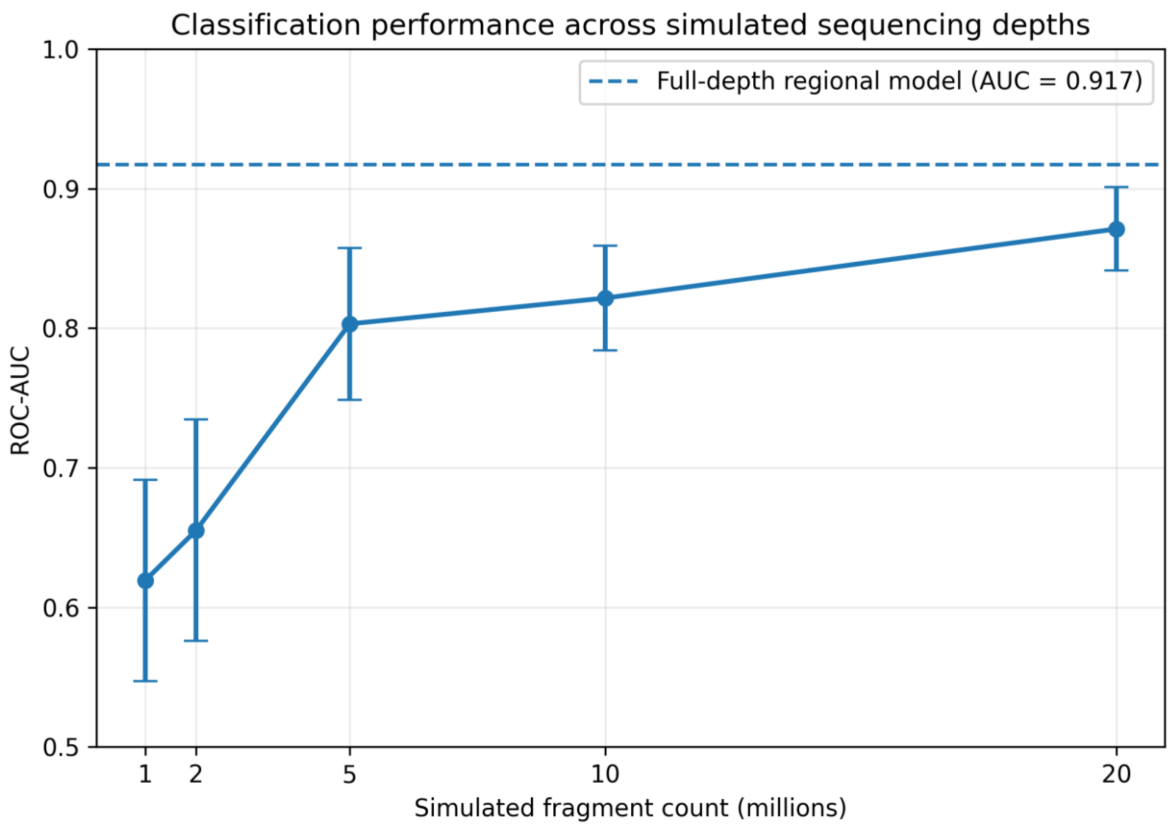

Figure 5. Classification performance across simulated sequencing depths. The dashed horizontal line indicates the full-depth regional-model ROC-AUC (0.917); error bars represent variability across 20 independent downsampling replicates.

Part 5. Interpretation & Biological Insight

Q10. Hypothesis Evaluation

The findings largely support the original hypothesis but also refine it. The first part of the hypothesis was that genome-wide regional fragmentation patterns would provide stronger discrimination between breast cancer and healthy samples than global fragment-size statistics. This was strongly supported. Global features produced a ROC-AUC of only approximately 0.65, whereas regional fragmentation achieved approximately 0.92.

This difference is biologically meaningful. A global feature such as mean fragment length reduces millions of cfDNA molecules to a single value. Even the global short-to-long ratio removes information about where those fragments originated in the genome. Regional profiles preserve spatial information and may therefore capture differences associated with chromatin accessibility, nucleosome organization, tissue of origin, and cancer-associated changes in DNA fragmentation. Fragmentomic studies have similarly demonstrated that genome-wide fragmentation patterns contain cancer-associated information that extends beyond individual sequence alterations (Cristiano et al., 2019).

The second part of the hypothesis was that diagnostic information would deteriorate as sequencing depth decreased. This was also supported. However, the decline was not immediate. At 20M fragments, regional profiles remained highly correlated with the full-depth profiles, and classification ROC-AUC remained approximately 0.87. At 10M and 5M fragments, useful classification signal was still present, although reduced. A much stronger deterioration occurred at 2M and 1M fragments.

The results therefore modify the original hypothesis by suggesting that regional fragmentomics has a depth-dependent operating range rather than a simple high-depth/low-depth boundary. Moderate reductions may retain useful biological information, whereas very low fragment counts no longer provide enough observations across genomic regions to reconstruct a stable fragmentation profile.

From an assay-development perspective, this is important because maximum sequencing depth is not necessarily the optimal design goal. Increasing depth adds cost, data storage, and computational burden. The more useful question is where additional sequencing stops providing enough additional information to justify the cost. These results suggest that the region between approximately 5M and 20M fragments deserves further investigation as a possible analytical operating range, but the present study is not sufficient to define a clinical minimum.

Q11. Proposed Wet-Lab Validation

A direct validation experiment would prospectively sequence plasma cfDNA from an independent cohort of patients with breast cancer and healthy controls. For each plasma sample, one sequencing library would be prepared and sequenced to sufficiently high depth. The same library could then be physically sequenced at different read allocations or computationally subsampled to predefined fragment counts of 20M, 10M, 5M, 2M, and 1M. Using the same library across depths would minimize variability from DNA extraction and library preparation while directly testing the effect of sequencing coverage.

Regional short-to-long fragmentation profiles would be generated at each depth and compared with the high-depth reference. The predefined classifier would then be applied without retraining on the validation cohort. Primary endpoints would include profile correlation, ROC-AUC, sensitivity, specificity, and reproducibility across technical replicates. This experiment would determine whether the depth-performance relationship observed computationally can be reproduced using independently generated clinical specimens.

Q12. Limitations and Dataset Bias

Several limitations should be considered. The cohort contained only 92 samples, which is relatively small for a machine-learning study involving hundreds of regional features. Although repeated stratified cross-validation improves the stability of the performance estimate, it does not replace independent external validation.

The dataset was also balanced intentionally between breast cancer and healthy controls. This is useful for method development but does not reflect real screening prevalence, where cancer is much less common. Therefore, the current results should not be interpreted as estimates of positive predictive value in a screening population.

Potential biological confounders such as age, cancer stage, subtype, treatment status, inflammatory conditions, and other comorbidities were not incorporated into the classifier. Fragmentation patterns can reflect multiple biological processes, so a cancer-versus-healthy signal does not necessarily mean every feature is tumor-specific.

In addition, computational downsampling reduces fragment availability but does not fully reproduce every effect of generating a genuinely lower-depth sequencing library. Library complexity, PCR duplication, GC bias, pre-analytical handling, extraction efficiency, and sequencing-platform effects may behave differently in real experiments.

Finally, the strong performance of hundreds of regional features relative to only 92 samples creates a risk of overfitting. Regularized logistic regression and cross-validation reduce this risk, but independent validation is required before the model could be considered clinically generalizable.

Part 6. Code Reproducibility

The complete analysis workflow, processed data required for downstream analysis, final figures, result tables, environment specification, and Python preprocessing code are available in the public GitHub repository:

GitHub repository: https://github.com/nelly-abualkheir/cfDNA-fragmentomics-depth

The repository contains a README.md describing the project and workflow, an environment.yml file defining the computational environment, the final Jupyter notebook, Python preprocessing code, processed cohort-level datasets, final result tables, and figures.

Large raw fragment files are not stored directly in GitHub. Instead, the repository documents the source of the public data and provides the computational workflow used to transform fragment-level data into the features used for downstream analysis. This keeps the repository practical while preserving reproducibility.

Conclusion

This project demonstrates that the diagnostic value of cfDNA fragmentomics depends not only on fragment size but also on genomic context and sequencing depth. Global fragment-size summaries showed limited discrimination between breast cancer and healthy samples, while genome-wide regional fragmentation patterns achieved substantially stronger classification performance. Moderate reductions in sequencing depth preserved part of this signal, but both profile stability and classification performance declined substantially at very low fragment counts.

The results support regional cfDNA fragmentation as a potentially informative signal for liquid-biopsy applications while emphasizing that sequencing depth should be treated as an analytical performance parameter. Future studies using larger independent cohorts and prospective sequencing experiments will be necessary to establish the minimum depth required for reproducible clinical performance.

References

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